



C5 Populations and communities (HL)

C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Populations in an environment with unlimited resources grow exponentially, but in real environments, density-dependent limiting factors (such as competition for food, predation, and disease) increasingly restrict growth as population size approaches the environment's carrying capacity (K), producing a logistic (S-shaped) growth curve. Density-independent factors (such as extreme weather) affect population size regardless of density. Interspecific interactions, including competition, predation, mutualism, herbivory and parasitism, structure communities; predator-prey relationships often show cyclical oscillations, with predator population peaks lagging behind prey population peaks because of the time needed for increased food availability to translate into increased predator survival and reproduction.

Worked example: follow the reasoning

A population of insects introduced to an island grows from 50 to 400 individuals over the first 10 weeks (roughly exponential), then growth slows and the population stabilises at approximately 900 individuals by week 30, remaining roughly constant afterward. Step 1: identify the exponential phase (weeks 0–10) where resources are abundant relative to population size and growth rate is high. Step 2: identify the deceleration phase (weeks 10–30) where density-dependent factors, such as increasing competition for limited food or space, progressively reduce the net growth rate. Step 3: identify the plateau (after week 30), where birth rate and death rate are approximately equal. Conclusion: the carrying capacity (K) of the island for this population is approximately 900 individuals, the population size at which resources balance losses from density-dependent limiting factors.

Quick check · multiple choice

In a logistic growth model, why does population growth rate slow as population size (N) approaches carrying capacity (K)?

- A Birth rate increases as N approaches K
- B Density-dependent factors, such as competition for resources, increasingly reduce birth rate and/or increase death rate
- C Immigration into the population stops completely
- D The mutation rate of the population decreases

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A predator population and its prey population in a nature reserve were monitored over 20 years. The prey population rose and fell in repeating cycles, and the predator population showed a similar cyclical pattern but consistently peaked 2–3 years after each prey peak. [7]
 - (a) Describe the relationship between the two population cycles shown by this data. [2]
 - (b) Explain why the predator population peak lags behind the prey population peak. [3]
 - (c) Suggest what happens to the prey population immediately after a predator population peak, and explain why. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) The predator and prey populations both show cyclical rises and falls over time, with the predator cycle following the same general pattern as the prey cycle but delayed, peaking roughly 2–3 years after each prey peak.

(b) When prey numbers rise, more food becomes available to predators, but it takes time for this extra food to be converted into increased predator survival, growth and reproduction (e.g. through gestation, offspring maturation); therefore the predator population only begins to increase substantially, and eventually peak, some time after the prey population has already peaked.

(c) The prey population declines sharply after a predator population peak, because the now-large predator population consumes prey faster than the prey population can reproduce/replace itself, reducing prey numbers; this prey decline will later cause a subsequent fall in the predator population once food becomes scarce, continuing the cycle.

Quick check answer

B. As population density increases, density-dependent factors like competition for food, space or mates intensify, lowering the birth rate and/or raising the death rate until growth rate approaches zero near K.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.